Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

If m is the root of the equation $(1-ab)x^2-\left(a^2+b^2\right)x-(1+ab)=0$ and m harmonic means are inserted between a and b, then the difference between the last and the first of the means equals

(A) $ab(a-b)$

(B) $a(b-a)$

(C) $ab(b-a)$

(D) $b-a$

Q.2

Given that for all $x \in \boldsymbol{R}$, $\quad \frac{1}{3}<\frac{x^2-2x+4}{x^2+2x+4}<3$, then $\frac{9.3^{2x}+6.3^x+4}{9.3^{2x}-6.3^x+4}$ lies between

(A) $\frac{1}{2}$ and 2

(B) 0 and 2

(C) $\frac{1}{3}$ and 3

(D) none of these

Q.3

$\left(x^2-4\right)$ is a factor of $f(x)=\left(a_1x^2+b_1x+c_1\right)\cdot\left(a_2x^2+b_2x+c_2\right)$ if

(A) $b_1=0, c_1+4a_1=0$

(B) $b_2=0, c_2+4a_2=0$

(C) $4a_1+2b_1+c_1=0, 4a_2+c_2=2b_2$

(D) $4a_1+c_1=2b_1, 4a_2+2b_2+c_2=0$

Q.4

The equation $x^{\frac{3}{4}(\log_2 x)^2+\log_2 x-\frac{5}{4}}=\sqrt{2}$ has

(A) atleast one real solution

(B) exactly three real solutions

(C) exactly one irrational solution

(D) complex roots

Q.5

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If $ax + by = 1, cx^2 + dy^2 = 1$ have only one solution, then

(A) $\frac{a^2}{c} + \frac{b^2}{d} = 1$

(B) $x = \frac{a}{c}$

(C) $y = \frac{b}{d}$

(D) none of these

Q.6

If $[.]$ represents greatest integer function, then $2[\mathrm{x}]^2 - 3[\mathrm{x}] + 1 = 0$ is true for $-$

(A) no real value of x

(B) $1 \leq x < 2$

(C) $x \in R$

(D) $x \in \left(\frac{\pi}{e}, 2\right)$

Q.7

If α, β are roots of the equation $\mathrm{ax}^2 + 3x + 2 = 0 (a < 0)$, then $\alpha^2/\beta + \beta^2/\alpha$ is less than-

(A) 1

(B) 2

(C) 0

(D) -1

Q.8

If α and β are roots of the equation $x^2 + px + q = 0$ and α^4 and β^4 are roots of $x^2 - rx + s = 0$, then choose the incorrect options for the roots of $x^2 - 4qx + 2q^2 - r = 0$

(A) one negative and one positive

(B) both positive

(C) both negative

(D) both real

Q.9

Passage:
If $(\mu) < 0$ is the necessary and sufficient condition for a particular real number μ to lie between the roots of a quadratic equation $f(x) = 0$, where $f(x) = ax^2 + bx + c$. Again if $f(\mu_1) f(\mu_2) < 0$, then exactly one of the roots will lie between μ_1 and μ_2.
Question:
If $|b| > |a + c|$, then $-$

(A) one root of $f(x) = 0$ is positive, the other is negative

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(B) exactly one of the roots of $f(x) = 0$ lies in $(-1, 1)$

(C) 1 lies between the roots of $\mathrm{f(x)} = 0$

(D) both the roots of $f(x) = 0$ are less than 1

Q.10

Passage:
If $(\mu) < 0$ is the necessary and sufficient condition for a particular real number μ to lie between the roots of a quadratic equation $f(x) = 0$, where $f(x) = ax^2 + bx + c$. Again if $f(\mu_1) f(\mu_2) < 0$, then exactly one of the roots will lie between μ_1 and μ_2.
Question:
If $a(a + b + c) < 0 < (a + b + c)c$, then $-$

(A) one root is less than 0, the other is greater than 1

(B) exactly one of the roots lies in $(0, 1)$

(C) both the roots lie in $(0, 1)$

(D) at least one of the roots lies in $(0, 1)$

Q.11

Passage:
If $(\mu) < 0$ is the necessary and sufficient condition for a particular real number μ to lie between the roots of a quadratic equation $f(x) = 0$, where $f(x) = ax^2 + bx + c$. Again if $f(\mu_1) f(\mu_2) < 0$, then exactly one of the roots will lie between μ_1 and μ_2.
Question:
If $(a + b + c)c < 0 < a(a + b + c)$, then $-$

(A) one root is less than 0 , the other is greater than 1

(B) one root lies in $(-\infty, 0)$ and other in $(0, 1)$

(C) both the roots lie in $(0, 1)$

(D) one root lies in $(0, 1)$ and other in $(1, \infty)$

Q.12

Passage:
Some equations can be solved by change of variables like
(i) The equation of the form $(x - a)(x - b)(x - c)(x - d) = A, a < b < c < d, b - a = d - c$ can be solved by change of variable $y = x - \frac{(a+b+c+d)}{4}$
(ii) The equation of the form $(x - a)(x - b)(x - c)(x - d) = Ax^2, ab = cd$, can be solved by change of variable $y = x + \frac{ab}{x}$
(iii) The equation of the form $(x - a)^4 + (x - b)^4 = A$, can be solved by change of variable $y = \frac{(x-a)+(x-b)}{2}$
Question:
Sum of all the real roots of the equation $(x - 1)(x - 3)(x - 5)(x - 7) = 9$ is -

(A) 4

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(B) 8

(C) 16

(D) None of these

Q.13

Passage:

Some equations can be solved by change of variables like

(i) The equation of the form $(x-a)(x-b)(x-c)(x-d)=A, a<b<c<d, b-a=d-c$ can be solved by change of variable $y=x-\frac{(a+b+c+d)}{4}$

(ii) The equation of the form $(x-a)(x-b)(x-c)(x-d)=Ax^2, ab=cd$, can be solved by change of variable $y=x+\frac{ab}{x}$

(iii) The equation of the form $(x-a)^4+(x-b)^4=A$, can be solved by change of variable $y=\frac{(x-a)+(x-b)}{2}$

Question:

Number of integral solution of the equation $(x+2)(x+3)(x+8)(x+12)=4x^2$ is

(A) 4

(B) 2

(C) 0

(D) None of these

Q.14

Passage:

Some equations can be solved by change of variables like

(i) The equation of the form $(x-a)(x-b)(x-c)(x-d)=A, a<b<c<d, b-a=d-c$ can be solved by change of variable $y=x-\frac{(a+b+c+d)}{4}$

(ii) The equation of the form $(x-a)(x-b)(x-c)(x-d)=Ax^2, ab=cd$, can be solved by change of variable $y=x+\frac{ab}{x}$

(iii) The equation of the form $(x-a)^4+(x-b)^4=A$, can be solved by change of variable $y=\frac{(x-a)+(x-b)}{2}$

Question:

Number of real roots of the equation $(3+x)^4+(5+x)^4=16$ is $-$

(A) 0

(B) 2

(C) 4

(D) 1

Q.15

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Match the column -

Column I	Column II
(A) If a, b, c and d are four non- zero real number such that $(d+a-b)^2+(d+b-c)^2=0$ and the roots of the equation $a(b-c)x^2+b(c-a)x+c(a-b)=0$ are real and equal, then	(p) $a+b+c=0$
(B) If the roots of the equation $(a^2+b^2)x^2-2b(a+c)x+(b^2+c^2)=0$ are real and equal, then	(q) a, b, c are in A.P.
(C) If the equation $ax^2+bx+c=0$ and $x^3-3x^2+3x-1=0$ have a common real root, then	(r) a, b, c are in G.P.
(D) Let a, b, c be positive real numbers such that the expression $bx^2+\sqrt{(a+c)^2+4b^2}\,(x)+(a+c)$ is non-negative $\forall\ x\in R$, then	(s) a, b, c are in H.P.

(A) (A) - q,s, (B) - p,s, (C) - q, (D) - r

(B) (A) - p, (B) - q,s, (C) - r, (D) - s

(C) (A) - q,r,s, (B) - r, (C) - p, (D) - q

(D) (A) - r,s, (B) - q, (C) - s, (D) - p

Q.16

If the equation $x^5-10a^3x^2+b^4x+c^5=0$ has three equal real roots, then the value of $6a^5+c^5$, is L and the value of $\frac{b^4}{a^4}$, is M. find the value of $\frac{L+M}{3}$ . . . . . .

Q.17

All the values of k for which the quadratic polynomial $f(x)=-2x^2+kx+k^2+5$ has two distinct zeroes and only one of them satisfying $0<x<2$, lie in the interval (a,b). Find the value of $(a+10\,b)$

Q.18

The integral value of λ for which the roots of equation $\lambda x^2+(2\lambda-1)x+\lambda-2=0$ are rational, if $2<\lambda\le 9$, is $-$

Q.19

The maximum possible number of real roots of equation $x^5-6x^2-4x+5=0$

Q.20

Let $\frac{\alpha}{\beta},\frac{\beta}{\alpha}$ be two roots of $(x+1)^n+x^n+1=0$ where $n>2, n\in N$, .Let α,β be roots of $x^2+px+q=0$. If α,β are also roots of $x^{2n}+p^nx^n+q^n=0$ then the smallest positive integral value of n is.

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Answers & Solutions

Q.1 Answer:

$ab(b-a)$

Solution:

By the given condition $(1-ab)m^2-\left(a^2+b^2\right)m-(1+ab)=0$

$\Rightarrow \quad m\left(a^2+b^2\right)+\left(m^2+1\right)ab=m^2-1$ **... (i)**

Now $H_1=$ **First H.M. between** a **and** $b=\frac{(m+1)ab}{a+mb}$ **and** $H_m=\frac{(m+1)ab}{b+ma}$

$$\therefore \quad H_m-H_1=(m+1)ab\left[\frac{1}{b+ma}-\frac{1}{a+mb}\right]=(m+1)ab\frac{[(m-1)(b-a)]}{(b+ma)(a+mb)}$$

$$=\frac{\left(m^2-1\right)ab(b-a)}{m\left(a^2+b^2\right)+\left(m^2+1\right)ab}=\frac{\left(m^2-1\right)ab(b-a)}{m^2-1} \qquad [\text{From (i)}]$$

$$=ab(b-a)$$

Q.2 Answer:

$\frac{1}{3}$ and 3

Solution:

$$\frac{9.3^{2x}+6\cdot 3^x+4}{9.3^{2x}-6\cdot 3^x+4}$$

$$=\frac{3^2\cdot 3^{2x}+2\cdot 3\cdot 3^x+4}{3^2\cdot 3^{2x}-2\cdot 3\cdot 3^x+4}$$

$$=\frac{\left(3^{x+1}\right)^2+2\cdot 3^{x+1}+4}{\left(3^{x+1}\right)^2-2\cdot 3^{x+1}+4}$$

$=\frac{t^2+2t+4}{t^2-2t+4}$, **where** $t=3^{x+1}$ **is real**

Since $\quad \frac{1}{3}<\frac{x^2-2x+4}{x^2+2x+4}<3$, **therefore** $\quad 3>\frac{x^2+2x+4}{x^2-2x+4}>\frac{1}{3}$

$\Rightarrow \quad \frac{1}{3}<\frac{x^2+2x+4}{x^2-2x+4}<3 \quad$ **for all** $x\in \boldsymbol{R}$.

Hence the given expression takes values in $\left(\frac{1}{3},3\right)$.

Q.3 Answer:

$b_1=0, c_1+4a_1=0$

Solution:

(a) x^2-4 **is a factor of** $a_1x^2+b_1x+c_1 \Rightarrow b_1=0, c_1+4a_1=0$

(b) x^2-4 **is a factor of** $a_2x^2+b_2x+c_2 \Rightarrow b_2=0, c_2+4a_2=0$

(c) $x-2$ **is a factor of** $a_1x^2+b_1x+c_1$ **and** $x+2$ **is a factor of** $a_2x^2+b_2x+c_2$

$\Rightarrow \quad 4a_1+2b_1+c_1=0$ **and** $4a_2-2b_2+c_2=0$

(d) $x+2$ **is a factor of** $a_1x^2+b_1x+c_1$ **and** $x-2$ **is a factor of** $a_2x^2+b_2x+c_2$

$\Rightarrow \quad 4a_1-2b_1+c_1=0$ **and** $4a_2+2b_2+c_2=0$

Q.4 Answer:

atleast one real solution

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Solution:

Let $\log_2 x = t$ then

$\frac{3}{4}\left(\log_2 x\right)^2 + \log_2 x - \frac{5}{4} = \frac{1}{2}\log_x 2$

$\Rightarrow \quad \frac{3}{4}t^2 + t - \frac{5}{4} = \frac{1}{2t} \Rightarrow 3t^2 + 4t - 5 = \frac{2}{t} \quad \Rightarrow 3t^3 + 4t^2 - 5t - 2 = 0$

A cubic equation has atleast one real solution, $t = 1$ is solution of the equation other values of $t = -2, -\frac{1}{3} \Rightarrow x = 2, \frac{1}{2^2}, \frac{1}{2^{1/3}}$ in which one is irrational.

Q.5 Answer:

$\frac{a^2}{c} + \frac{b^2}{d} = 1$

Solution:

We have, $ax + by = 1 \Rightarrow y = \frac{1-ax}{b}$

Putting this value in the second equation, we get

$$cx^2 + \frac{d}{b^2}(1 - ax)^2 = 1$$

$$\Rightarrow \quad \left(b^2c + a^2d\right)x^2 - 2adx + d - b^2 = 0 \quad \ldots(i)$$

This quadratic equation will have equal roots if

$$D = 4a^2d^2 - 4\left(b^2c + a^2d\right)\left(d - b^2\right) = 0$$

$$\Rightarrow \quad a^2d^2 + \left(b^2c + a^2d\right)b^2 - b^2cd - a^2d^2 = 0 \quad \Rightarrow \quad b^2\left(b^2c + a^2d - cd\right) = 0$$

$$\Rightarrow \quad b^2c + a^2d = cd \Rightarrow \frac{b^2}{d} + \frac{a^2}{c} = 1$$

$$\text{Also, } x = \frac{2ad}{2\left(b^2c + a^2d\right)} = \frac{ad}{cd} = \frac{a}{c}$$

$y = \frac{1-ax}{b} = \frac{1}{b}\left(1 - \frac{a^2}{c}\right) = \frac{1}{b} \cdot \frac{b^2}{d} = \frac{b}{d}$

Q.6 Answer:

$1 \leq x < 2$

Solution:

$$2[x]^2 - 3[x] + 1 = 0$$

$$\text{Let } [x] = y$$

$$2y^2 - 3y + 1 = 0$$

$$(2y - 1)(y - 1) = 0$$

$$y = 1 \text{ or } y = 1/2 \notin 1$$

$\therefore \quad [x] = 1$

$\therefore \quad 1 \leq x < 2$

Q.7 Answer:

1

Solution:

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Since $a < 0$, therefore discriminant

$D = 9 - 8a > 0$. **So, α and β are real. We have :** $\alpha + \beta = \frac{-3}{a}$ **and** $\alpha\beta = \frac{2}{a}$

$\therefore \quad \frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} = \frac{\alpha^3+\beta^3}{\alpha\beta} = \frac{(\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)}{\alpha\beta} = \frac{(\alpha+\beta)^3}{\alpha\beta} - 3(\alpha+\beta) = -\frac{27}{2a^2} + \frac{9}{a} < 0 [a < 0]$

Q.8 Answer:

both positive

Solution:

$$\begin{aligned} \text{The product of roots } &= 2q^2 - r \\ &= 2(\alpha\beta)^2 - \left(\alpha^4 + \beta^2\right) \\ &= -\left(\alpha^2 + \beta^2\right) < 0 \end{aligned}$$

As the product of roots is < 0, so one root has to be negative and the other has to be positive.

Q.9 Answer:

exactly one of the roots of $f(x) = 0$ lies in $(-1, 1)$

Solution:

(i) $b^2 > (a+c)^2$

$\Rightarrow (a+c-b)(a+c+b) < 0 \Rightarrow f(-1)f(1) < 0$

So, there is exactly one root in $(-1, 1)$.

(ii) af $(1) < 0$ **and** $f(0)f(1) > 0$

$\Rightarrow$ **af** $(1) < 0$ **and af** $(0) < 0$

Hence, both the numbers 0 and 1 lie between the roots.

(iii) $f(0)f(1) < 0$ **and af** $(1) > 0$

$\Rightarrow f(0)f(1) < 0$ **and af** $(0) < 0$

Hence, exactly one root lies in $(0, 1)$ **and 0 lies between the roots.**

Q.10 Answer:

both the roots lie in $(0, 1)$

Solution:

(i) $b^2 > (a+c)^2$

$\Rightarrow (a+c-b)(a+c+b) < 0 \Rightarrow f(-1)f(1) < 0$

So, there is exactly one root in $(-1, 1)$.

(ii) af $(1) < 0$ **and** $f(0)f(1) > 0$

$\Rightarrow$ **af** $(1) < 0$ **and af** $(0) < 0$

Hence, both the numbers 0 and 1 lie between the roots.

(iii) $f(0)f(1) < 0$ **and af** $(1) > 0$

$\Rightarrow f(0)f(1) < 0$ **and af** $(0) < 0$

Hence, exactly one root lies in $(0, 1)$ **and 0 lies between the roots.**

Q.11 Answer:

one root lies in $(-\infty, 0)$ and other in $(0, 1)$

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Solution:

(i) $b^2 > (a+c)^2$

$\Rightarrow (a+c-b)(a+c+b) < 0 \Rightarrow f(-1)f(1) < 0$

So, there is exactly one root in $(-1, 1)$.

(ii) af $(1) < 0$ **and** $f(0)f(1) > 0$

$\Rightarrow$ **af** $(1) < 0$ **and af** $(0) < 0$

Hence, both the numbers 0 and 1 lie between the roots.

(iii) $\mathrm{f}(0)\mathrm{f}(1) < 0$ **and af** $(1) > 0$

$\Rightarrow \mathrm{f}(0)\mathrm{f}(1) < 0$ **and af** $(0) < 0$

Hence, exactly one root lies in $(0, 1)$ **and 0 lies between the roots.**

Q.12 Answer:

16

Solution:

Here $3 - 1 = 7 - 5$

$\therefore$ **make the substitution** $\mathrm{y} = \mathrm{x} - \frac{1+3+5+7}{4} = \mathrm{x} - 4$

$\therefore$ $\mathrm{y}^2 = 0, 10$ **i.e.,** $\mathrm{y} = 0, 0, -\sqrt{10}, \sqrt{10}$

$\therefore$ **roots of the original are** $x = 4, 4, 4 - \sqrt{10}, 4 + \sqrt{10}$

$\therefore$ **sum of roots is 16.**

Since $(-2)(-12) = (-3)(-8)$ **so we can introduce a new variable.**

$\mathrm{y} = \mathrm{x} + \frac{24}{\mathrm{x}}$ **after solving we get** $\mathrm{x} = \frac{-15 \pm \sqrt{129}}{2}, -6, -4$ **so number of integral roots is 2.**

Put $y = x + 4 \Rightarrow x = y - 4$**, we get** $x = -5$ **or** -3

so number of real roots is 2.

Q.13 Answer:

2

Solution:

Here $3 - 1 = 7 - 5$

$\therefore$ **make the substitution** $\mathrm{y} = \mathrm{x} - \frac{1+3+5+7}{4} = \mathrm{x} - 4$

$\therefore$ $\mathrm{y}^2 = 0, 10$ **i.e.,** $\mathrm{y} = 0, 0, -\sqrt{10}, \sqrt{10}$

$\therefore$ **roots of the original are** $x = 4, 4, 4 - \sqrt{10}, 4 + \sqrt{10}$

$\therefore$ **sum of roots is 16.**

Since $(-2)(-12) = (-3)(-8)$ **so we can introduce a new variable.**

$\mathrm{y} = \mathrm{x} + \frac{24}{\mathrm{x}}$ **after solving we get** $\mathrm{x} = \frac{-15 \pm \sqrt{129}}{2}, -6, -4$ **so number of integral roots is 2.**

Put $y = x + 4 \Rightarrow x = y - 4$**, we get** $x = -5$ **or** -3

so number of real roots is 2.

Q.14 Answer:

2

Solution:

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Here $3-1=7-5$

$\therefore$ make the substitution $y=x-\frac{1+3+5+7}{4}=x-4$

$\therefore$ $y^2=0,10$ i.e., $y=0,0,-\sqrt{10},\sqrt{10}$

$\therefore$ roots of the original are $x=4,4,4-\sqrt{10},4+\sqrt{10}$

$\therefore$ sum of roots is 16.

Since $(-2)(-12)=(-3)(-8)$ so we can introduce a new variable.

$y=x+\frac{24}{x}$ after solving we get $x=\frac{-15\pm\sqrt{129}}{2},-6,-4$ so number of integral roots is 2.

Put $y=x+4\Rightarrow x=y-4$, we get $x=-5$ or -3

so number of real roots is 2.

Q.15 Answer:

(A) - q,r,s, (B) - r, (C) - p, (D) - q

Solution:

$(A)-qrs,(B)-r,(C)-p,(D)-q$

$d+a-b=0$ and $d+b-c=0$

$d=b-a$ and $d=c-b$

$\therefore$ $b-a=c-b\Rightarrow 2b=a+b\Rightarrow a,b,c$ are in A.P.

Also, $x=1$ satisfies the second equation.

Therefore, the other root is also 1. Product of root is 1.

$\therefore$ $c(a-b)=a(b-c)\Rightarrow b=\frac{2ac}{a+c}\Rightarrow a,b,c$ are in H.P.

Therefore, a,b,c are in A.P. and a,b,c in H.P. Hence a,b,c are in G.P.

(B) $\left(a^2+b^2\right)x^2-2b(a+c)x+\left(b^2+c^2\right)=0$

The roots are real and equal. Hence,

$$4b^2(a+c)^2-4\left(a^2+b^2\right)\left(b^2+c^2\right)=0$$

$$\Rightarrow b^2\left(a^2+c^2+2ac\right)-\left(a^2b^2+a^2c^2+b^4+b^2c^2\right)=0$$

$$\Rightarrow b^2a^2+b^2c^2+2ab^2c-a^2b^2-a^2c^2-b^4-b^2c^2=0$$

$$\Rightarrow 2ab^2c-a^2c^2-b^4=0\Rightarrow\left(b^2-ac\right)^2=0$$

Hence, $b^2=ac$. Thus, a,b,c are in G.P.

(C) $(x-1)^3=0\Rightarrow x=1$ is the common root. Hence, $a+b+c=0$.

(D) $(a+c)^2+4b^2-4b(a+c)\leq 0,x\in R$

$\Rightarrow$ $((a+c)-2b)^2\leq 0\Rightarrow a+c=2b\Rightarrow a,b,c$ in A.P.

Q.16 Answer:

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Solution:

$x^5-10a^3x^2+b^4x+c^5=f(x)$

$f^1(x)=5x^4-20a^3x+b^4$ which implies $x=a$ is three times repeated root

$f^{11}(x)=20x^3-20a^3=0\Rightarrow x=a$

$f^1(a)=5a^4-20a^4+b^4=0\Rightarrow -15a^4+b^4=0\Rightarrow\frac{b^4}{a^4}=15$

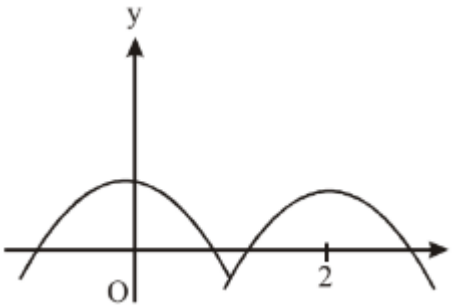
$a^5-10a^5+b^4a+c^5=f(a)=0\&b^4=15a^4\Rightarrow 6a^5+c^5=0$

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Q.17 Answer:

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Solution:



For two distinct roots, $\mathrm{D} > 0$ **i.e.** $\mathrm{k}^2 + 8\left(\mathrm{k}^2 + 5\right) > 0$ **which is always true**

Also let

$f(x) = -2x^2 + kx + k^2 + 5 = 0$

$\mathrm{f}(0)$ **and** $\mathrm{f}(2) < 0$ **or** $\mathrm{f}(0) < 0$ **and** $\mathrm{f}(2) < 0$

hence, $\mathrm{f}(0) \cdot \mathrm{f}(2) < 0$

but $\mathrm{f}(0) = \mathrm{k}^2 + 5 > 0$ **always true**

gives $f(2) < 0$

$-8 + 2\mathrm{k} + \mathrm{k}^2 + 5 < 0 \Rightarrow \mathrm{k}^2 + 2\mathrm{k} - 3 < 0 \Rightarrow (\mathrm{k} + 3)(\mathrm{k} - 1) < 0$

$\mathrm{k} \in (-3, 1) \Rightarrow \mathrm{a} = -3,\ \mathrm{b} = 1 \Rightarrow \mathrm{a} + 10\,\mathrm{b} = -3 + 10 = 7$

Q.18 Answer:

—

Solution:

$\lambda x^2 + (2\lambda - 1)x + \lambda - 2 = 0$

$D = (2\lambda - 1)^2 - 4\lambda(\lambda - 2) = 4\lambda + 1$ **(an odd number)**

$\Rightarrow D$ **must be perfect square for rational roots**

$4\lambda + 1 = (2k + 1)^2 \quad k \in I$

$\lambda = k(k + 1)$

so $\lambda = 0, 2, 6, 12$

$\lambda = 6$

Q.19 Answer:

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Solution:

$f(x) = x^5 - 6x^2 - 4x + 5 = 0$

$= +\quad -\quad -\quad +$

Number of change of sign $= 2$

Atmost 2 positive real roots

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$f(-x) = -x^5 - 6x^2 + 4x + 5 = 0$



Number of change of sign $= 1$

Atmost 1 negative real roots

OR

$+$ ve real root (max.)	$-$ ve real root (max.)	Imaginary root
2	1	2

$\Rightarrow (\text{Max})$ **real roots** $= 3$

Q.20 Answer:

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Solution:

As $\frac{\alpha}{\beta}$ **is a root of** $(x+1)^n + x^n + 1 = 0$**, we get** $\left(\frac{\alpha}{\beta}+1\right)^n + \left(\frac{\alpha}{\beta}\right)^n + 1 = 0 \Rightarrow (\alpha+\beta)^n + \alpha^n + \beta^n = 0.$

But α, β **are roots of** $x^2 + px + q = 0 \Rightarrow \alpha + \beta = -p, \alpha\beta = q \Rightarrow (-p)^n + \alpha^n + \beta^n = 0$ **As** α, β **are roots of** $x^{2n} + p^n x^n + q^n = 0$**, We get** $\alpha^{2n} + p^n\alpha^n + q^n = 0$ **and similarly for** β**.**

Subtracting these two and with simplification we get $\alpha^n + \beta^n + p^n = 0.$

So, finally $(-p)^n = p^n$ **so** n **is an even number**

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